

Discriminative Project

CNN / Deep Learning Based Automated Celebrity Face Detection System

Milestones 1 & 2

Group 02

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Git Repo Link: github.com/AHonGa/ie7615-group-02-project1

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Work Summary & Key Insights

Milestone 1 — Classification Baseline

We built and compared two classifiers on a 4-identity CelebA subset (identities 3, 7, 1212, 8335 — the IDs individually claimed by each team member for the class-wide identity-pooling task, 23-25 images each): a custom CNN trained from scratch, and a ResNet18 fine-tuned from ImageNet-pretrained weights.

Model	Test Acc.	Val Acc.	Params	Train Time (s)
ResNet18 (fine-tuned)	0.875	0.9375	11,178,564	461.3
Custom CNN (from scratch)	0.500	0.6875	1,207,588	927.1

Key insights:

(1) With only ~17 training images per class, a pretrained backbone's transfer-learning advantage is decisive — ResNet18 outscored the from-scratch CNN by 37.5 points, a much wider gap than on an easier, more visually diverse identity set (where the gap was only 10 points), because pretrained ImageNet features already encode general-purpose visual structure the custom CNN cannot learn from so little data. (2) Hyperparameter sensitivity is real for from-scratch models: an initial custom-CNN configuration (heavier augmentation, higher learning rate) collapsed to predicting a single class; lightening augmentation, lowering the learning rate, and reducing dropout fixed this and produced a genuine, if modest, 50% test accuracy. (3) Identity choice materially affects difficulty — our updated 4-identity subset (3 of 4 male, overlapping hair-color attributes) proved harder to separate than an independently diversity-selected 5-identity set used earlier, lowering both models' accuracy but especially the custom CNN's.

Milestone 2 — Detection Dataset Construction

We built a synthetic multi-celebrity detection dataset: 6 composite 3x3 grid images (640x640, YOLOv8's default input size), each combining 9 unique identities drawn from the class-wide identity pool (13 identities across all 5 groups, not just our own team's 4), with exact YOLO-format bounding box annotations generated directly from the placement coordinates used to compose each grid.

Split	Base grids	Total images
Train	4	20
Validation	1	5
Test	1	5

Key insights:

(1) Generating bounding boxes directly from the composition step (rather than manually annotating afterward) eliminates a whole class of labeling error, since each box is exact by construction. (2) Detection-appropriate augmentation is not just image transforms — every transform (flip, rotation, zoom) had to be paired with a corresponding coordinate transform on its bounding box, or the labels silently become wrong; we verified this held under real rotation/zoom by rendering original-vs-augmented images with boxes drawn on both. (3) With so few base images (6 grids), naively splitting by individual image after augmentation would leak the same underlying grid across train and test; splitting at the base-grid level before augmenting was necessary to get a genuinely leak-free evaluation. (4) Drawing identities from the whole class pool (not just our own team's 4) required coordinating with other groups via a shared spreadsheet — a preview of the cross-team coordination Milestone 3's aggregated evaluation will also require.

Looking Ahead to Milestone 3

The dataset produced in Milestone 2 (`detection_dataset/{train,val,test}/`) is the direct input to Milestone 3, where we will fine-tune YOLOv8 on these images and labels, tune detection hyperparameters (learning rate, confidence threshold, NMS IoU threshold, image size, augmentation intensity), and report mAP@0.5, mAP@0.5:0.95, IoU, precision, and recall.